

Advanced Warehouse Management In D365 Finance and Operations

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Introduction

Microsoft Dynamics 365 Finance and Operations (D365 F&O) offer advanced warehouse management capabilities that aid businesses in improving their warehouse processes. This guide provides detailed steps for setting up and managing an advanced warehouse in D365 F&O.

This document is created with assumption that a warehouse has 5 inbound staging locations, 5 outbound staging location and rest of the warehouse area is floor used to store inventory in aisle, shelf, rack and bin. The actual inventory is not stored on License pallets.

Step-by-Step Guide

1. Create a Storage Dimension Group

To create a storage dimension group:

- Navigate to Product information management > Setup > Dimension & variant groups > Storage dimension groups.
- Click New to create a new storage dimension group.
- Enter a Name and Description.
- Select the necessary storage dimensions such as Site, Warehouse, Location, and others as per image below.
- Save the storage dimension group.

Note: when you select use warehouse management processes to yes, system by default enables License Plate in setup.

The screenshot displays the 'Storage dimension groups' setup page in Microsoft Dynamics 365 Finance and Operations. The breadcrumb navigation at the top reads: 'Finance and Operations > Product information management > Setup > Dimension and variant groups > Storage dimension groups'. The user is logged in as 'USMF'.

Storage dimension groups

Storage dimensions

Name	Active	Blank recei...	Blank issue...	Physical in...	Financial i...	Coverage ...	For purcha...	For sales p...
Site	✓			✓	✓	✓	✓	✓
Warehouse	✓			✓		✓	✓	✓
Location	✓			✓				
Inventory status	✓			✓				
License plate	✓	✓	✓	✓				

Warehouse specific setup

Use warehouse management pro... ☒ Yes

WAREHOUSE SPECIFIC SETUP

Mandatory ☐ No

Primary stocking ☒ Yes

2. Create a Tracking Dimension Group

To create a tracking dimension group:

- Navigate to Product information management > Setup > Dimension and variant groups > Tracking dimension groups.
- Click New to create a new tracking dimension group.
- Enter a Name and Description.
- Select the necessary tracking dimensions such as Batch number, Serial number, etc. as per image below.
- Save the tracking dimension group.

Name	Active	Active in sales process	Primary st...	Blank recei...	Blank issue...	Physical in...	Financial i...	Coverage ...	For purcha...	For sales p...	Transfer
Batch nu...	☒					☒	☒				
Serial nu...											
Owner						☒	☒				☒

3. Create a Warehouse and assign a name and site

To create a warehouse:

- Navigate to Warehouse management > Setup > Warehouse > Warehouses.
- Click New to create a new warehouse.
- Enter a Warehouse ID, Name, and assign a Site.
- Save the warehouse.

Warehouse	Name
80	Burlington

General			
Site	Quarantine warehouse	Goods in transit warehouse	REFERENCE
2	28	24-GIT	Vendor account
Type	Transit warehouse	Under delivery warehouse	
Default	27	24-UND	

Master planning			
Inventory and warehouse management			
Default receipt location	Default production input location	Default shipment maintenance lo...	Work processing policy name
Default issue location	Issue status after raw material pi...	Default location for credit only re...	Default quality maintenance loca...
	Picked		
Default production finished goo...	Default kanban finished goods lo...	Default container type	Default location for final shipment
			Default location for sorting

4. Assign Warehouse Management Attributes

To assign warehouse management attributes:

- Navigate to the Warehouse created in the earlier step.
- Under the General tab, select the necessary attributes such as Use warehouse management processes.
- Save the changes.

Standard view ▾

Warehouses

Warehouse: 80 Name: Burlington

Hierarchy: Retail

Warehouse

Use warehouse management processes: ☒ Yes

Requirement for inventory reservation: Allow partial reservation ▾

REDUCE PICKED QUANTITY

Decrement load line: ☒ No

Enable quality order for warehouse: ☒ No

Print BOL before confirming shipment: ☒ No

Release rule failure option for audit: Warning and skip line ▾

INVENTORY

Default inventory status ID: ▾

Default container type: ▾

Work processing policy name: ▾

Auto update shipment: On quantity decrease ▾

RESERVATIONS

Reserve only warehouse in production: ☒ No

Reserve inventory at load posting: ☒ No

NEGATIVE INVENTORY

Financial negative inventory: ☒ No

Physical negative inventory: ☒ No

CYCLE COUNTING

Allow license plate moves during: ☒ No

INVENTORY STATUS CHANGE

Remove reservations and markings: None ▾

Remove inventory blocking: ☒ No

PRODUCTION ORDERS

Backflush must use reserved quantity: ☒ No

Posting method: Inherit ▾

HAZARDOUS GOODS REPORT LANGUAGE

Hazardous goods domestic report: ▾

Hazardous goods export report: ▾

5. Create Location Profiles

To create a location profile:

- Navigate to Warehouse management > Setup > Warehouse > Location Profiles.
- Click New to create a new location format.
- Enter a Location Profile ID and Name.
- Define the location format and location type and other attributes as per image below
- Save the location profile.
- Likewise create multiple location profiles as per requirement.

Finance and Operations Warehouse management > Setup > Warehouse > Location profiles USMF

Save + New Delete Options

Standard view ▾

Location profiles

Location profile ID: InboundStage Name: Inbound Stage

General

Location format: STAGE ▾

Location type: STAGE ▾

Dock management profile ID: ▾

Use license plate tracking: ☒ No

Allow negative inventory: ☒ Yes

Allow mixed items: ☒ Yes

Enable location product dimensions: ☒ No

Allow mixed inventory statuses: ☒ No

Allow mixed inventory batches: ☒ Yes

Allow cycle counting: ☒ Yes

Generate check digit: ☒ No

Override rules for batch days: ☒ No

Allowed container type group: ▾

Enable item in location: ☒ No

Enable location activity date and time: ☒ No

Enable location status: ☒ No

Enable license plate positioning: ☒ No

Display mobile device LP position: ☒ No

6. Create Locations

To create locations formats and add segments:

- Navigate to Warehouse management > Setup > Warehouse > Locations.
- Click location setup wizard.
- Select warehouse, location profile id
- In lines define from number and To number.
- Click create.

Standard view

Location setup wizard

Warehouse

Location profile ID

Additional Zone 2

80

InboundStage

Zone ID

Additional Zone 1

Additional Zone 3

		Line number ↑	Segment description	From num...	To number	Static text	Length
		1.0000000000	Stage	1	5		10

Create

Cancel

Like wise create location for Outbound staging area as well as Storage floor.

7. Create Inbound Location Directives

To create inbound location directives:

- Navigate to Warehouse management > Setup > Location directives.
- Click New to create a new location directive.
- Select Inbound as the directive type.
- Define the criteria and actions for the location directive.
- Save the location directive.

Note: screenshot added in step for adding location directive.

8. Add Lines to the Location Directive

To add lines to the location directive:

- Navigate to the created location directive.
- Under the Lines tab, click Add line.
- Define the criteria for the line.
- Save the changes.

9. Add Location Directive Actions to the Location Directive

To add location directive actions:

- Navigate to the created location directive.
- Under the Actions tab, click Add action.
- Define the actions for the location directive.
- Save the changes.

The screenshot shows the 'Location directives' setup screen. On the left, a list of location directives is shown, with 'Burlington Pick' selected. The main area displays the configuration for 'Burlington Pick'. The 'Work type' is 'Pick', the 'Directive code' is empty, and the 'Scope' is 'Single item'. The 'Warehouse selection' section shows 'Warehouse' set to '80'. The 'Lines' table has two rows:

Sequence	From quantity	To quantity	Unit	Locate quantity	Restrict by	Round up to	Locate package	Allow split	Immediate repl
1	0.00	10,000.00	BALE	None	✓				
2	0.00	10,000.00	BALE	None				✓	

Note: In Pick no location directive needs to be added. In put location directive is needed. Update query as needed.

Similarly create Put single, put multiple and inbound location directives.

10. Create Work Template

To create a work template:

- Navigate to Warehouse management > Setup > Work > Work templates.
- Click New to create a new work template.
- Enter the necessary details for the work template.
- Save the work template.

Finance and Operations Warehouse management > Setup > Work > Work templates USMF

Work templates

Standard view *~

Work order type
Sales orders

Overview General

Sequence	Work template	Work template description	Valid	Automatic...	Work pool ID	Work prior...
0	Burlington SO Process	Ship direct	☒	☐		0

Work Template Details

+ New Delete Move up Move down Work line breaks

Line number	Work type	Custom work type	Mandatory	Stop work	Freeze	Directive code	Work class ID	Quality check temp...	Anchoring gro
1.0000000000	Pick		☒	☐	☐		Sales		
2.0000000000	Put		☒	☐	☐	ToOutLoc	Sales		

11. Create Wave Template

To create a wave template:

- Navigate to Warehouse management > Setup > Waves > Wave templates.
- Click New to create a new wave template.
- Enter the necessary details for the wave template.
- Save the wave template.

Finance and Operations Warehouse management > Setup > Waves > Wave templates USMF

Wave template

Wave template name Wave template description Wave template sequence Template status

Burlington Shipping Burlington Shipping 1 Valid

Warehouse selection Warehouse 80

Warehouse selection Site Warehouse

Warehouse 2 80

General

WAVE TEMPLATE INFORMATION

Automate wave creation ☒ Yes

Assign to open waves ☐ No

Process wave at release to wareh... ☒ Yes

Process wave automatically at th... ☐ No

Automate wave release ☒ Yes

Automate replenishment work re... ☒ Yes

Continue wave processing when ... ☐ No

WAVE THRESHOLDS

Wave weight threshold 0.00

Shipment threshold 0

Line threshold 0

DEFAULT VALUES

Attribute 1

Attribute 2

WAVE EXECUTION NOTIFICATIONS

Wave notification policy

Methods

12. Configure warehouse management parameters

To configure warehouse management parameters:

- Navigate to Warehouse management > Setup > warehouse management parameters
- Update location types as per requirement

Finance and Operations | Warehouse management > Setup > Warehouse management parameters | USMF

Standard view

Warehouse management parameters

General

Set up parameters for Warehouse management

Company information

Location profiles

User Location Profile: USER

Location types

Staging location type: STAGE | Packing location type: FLOOR | Final shipping location type: STAGE | Sorting location type:

Receiving

Exception for missing items from...: ASN | Transfer order receiving process: Combine the registration a... | Enable license plate receiving his...: No

- Configure wave parameters as per requirement

Finance and Operations | Warehouse management > Setup > Warehouse management parameters | USMF

Standard view

Warehouse management parameters

General

Set up parameters for Warehouse management

Wave processing

Enable schedule work creation: ☒

Wave processing batch group:

Process waves in batch: ☒

Create wave progress log: ☒ Yes

Create wave processing history log: ☒ Yes

Create containerization history log: ☒ Yes

Wait for lock (ms): 60000

Disable wave processing batch n...: ☒ No

Put location concurrency strictness: Performance optimized

Wait for put location directive lo...: 0

Ensure other parameters are also set correctly,

13. Update default locations on warehouse

To update default location for receipt and final location,

- Navigate to Warehouse management > Setup > Warehouse > Warehouses.
- Filter required warehouse
- Navigate to inventory and warehouse management tab and update location

14. Create a Catch Weight Handling Policy

To create a catch weight handling policy:

- Navigate to Product information management > Setup > Catch weight policies.
- Click New to create a new catch weight handling policy.
- Enter a Name and Description.
- Define the settings for the policy.
- Save the catch weight handling policy.

15. Create a New Released Product with catchweight attributes

To create a new released product:

- Navigate to Product information management > Products > Released products.
- Click New to create a new released product.
- Enter the necessary product details.
- Assign the right storage and tracking dimension groups.
- Save the released product.

16. Create Purchase Order and Receive It

To create a purchase order:

- Navigate to Procurement and sourcing > Purchase orders > All purchase orders.
- Click New to create a new purchase order.
- Enter the vendor and product details.
- Confirm the purchase order and receive the items.

17. Create Sales Order and Process It via Advanced Warehouse Management

To create a sales order:

- Navigate to Sales and marketing > Sales orders > All sales orders.
- Click New to create a new sales order.
- Enter the customer and product details.
- Confirm the sales order and process it using the advanced warehouse management features.

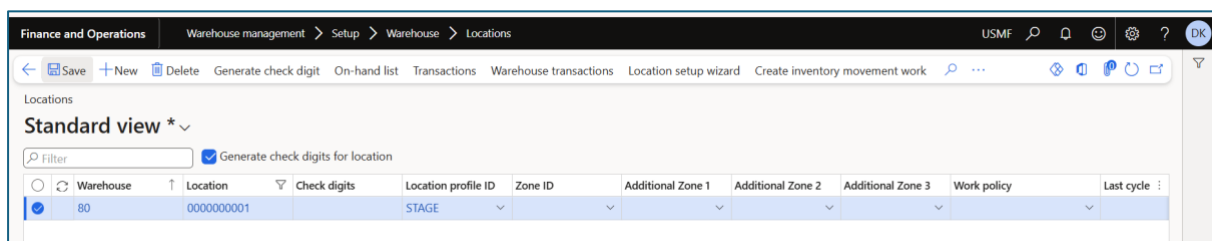
Conclusion

Setting up and managing an advanced warehouse in D365 Finance and Operations involves several detailed steps that ensure the warehouse functions efficiently. By following this guide, you can streamline your warehouse processes and improve overall productivity.

Important Notes:

By design advance warehouse functionality is enabled, in storage dimensions license plates are enabled by default. Using this guide we can store inventory without license plates (LP).

However, system requires LP for processing transaction. Hence to successfully process transactions, we have designated one location on inbound staging area and one location at outbound staging area as pallet enabled location. In order to consumes pallet at put locations, we have automated pallet creation in warehouse device using below showcased out of the box functionalities.



Warehouse	Location	Check digits	Location profile ID	Zone ID	Additional Zone 1	Additional Zone 2	Additional Zone 3	Work policy	Last cycle
80	0000000001	STAGE							

Standard view

Mobile device menu items

Menu item name: Sales Picking Title: Sales Picking Mode: Work Use existing work: ☒ Yes

General

Display inventory status: ☒ Yes	Anchoring: ☐ No	Count total item quantity first: ☐ No	Allow over pick: ☒ Yes
Directed by: User directed	Override target license plate: ☐ No	Disable cycle count thresholds: ☐ No	Group pick: ☐ No
Use process guide: ☐ No	Override license plate during put: ☐ No	Group put away: ☐ No	Deferred put processing policy: Respect settings
Barcode data policy:	Allow splitting of work: ☐ No	Pick oldest batch: None	Fast validation: ☐ No
Pick and pack: ☐ No	Keep work locked by user: ☐ No	Audit template ID:	Show work line list: Never show
Generate license plate: ☒ Yes	Display summary of pick screen: ☐ No	Handle by license plate: ☐ No	

Work classes

+ New Delete

Finance and Operations Warehouse management > Setup > Mobile device > Mobile device menu items USMF

Standard view

Mobile device menu items

Menu item name: Purchase Receive Title: Purchase Receive Mode: Work Use existing work: ☐ No

General

Use default data: ☐ No	License plate grouping policy: None	Display batch disposition code: ☐ No	Generate catch weight tag: ☒ Yes
Work creation process: Purchase order line receiving	Generate license plate: ☒ Yes	Print label: ☐ No	Assign putaway cluster: ☐ No
Use process guide: ☐ No	Work template:	Block receiving with ASN item in...: ☐ No	Deferred receiving policy ID:
Barcode data policy:	Display disposition code: ☐ No	Display container type: ☐ No	

Inventory reservation:

System is configured to consume inventory on FIFO basis, if the inventory on location is partially available, system will automatically consume inventory from next available location and create pick work transaction fro same.